Industrial Internship Report on

”Weather Monitoring System Using Node-RED and MQTT(Without Hardware)”

Prepared by

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Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time. My project was Weather Monitoring System using Node-RED and MQTT(Without Hardware). The project generates simulated temperature, humidity and weather- status data. The generated data is formatted in JSON and Debug notes were used to process and monitor the data. The processed information is finally displayed on a Node-RED dashboard using gauges and text/status widgets. This internship gave me a very good opportunity to get exposure to Industrial problems and design/ implement solution for that. It was an overall great experience to have this internship.

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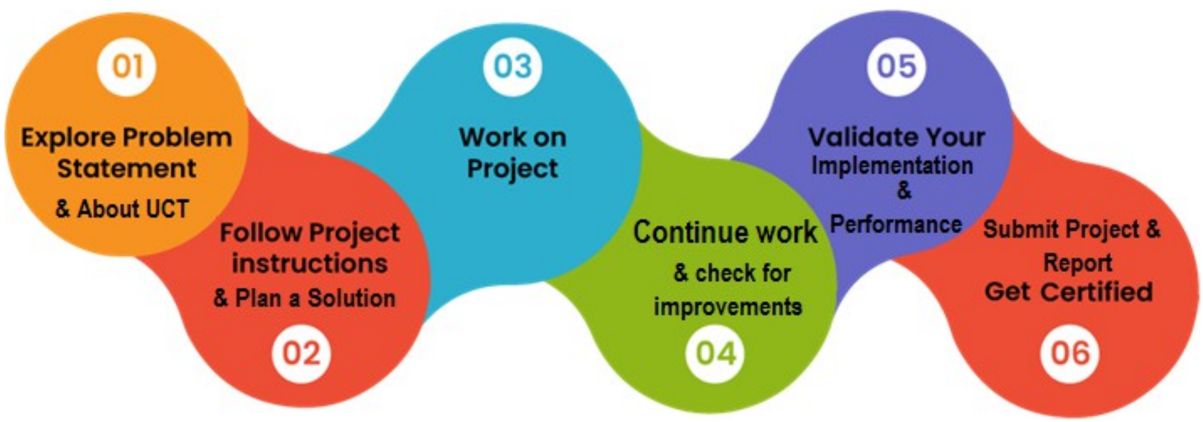
1. Preface

During the six-week internship, I learned IoT fundamentals, architecture, applications, MQTT, Node-RED and JSON data processing. I developed a **Weather Monitoring System without hardware**, created a Node-RED dashboard, implemented MQTT communication and practiced debugging and troubleshooting.

The project "**Weather Monitoring System using Node-RED and MQTT(Without Hardware)**" monitors simulated temperature, humidity and weather-status data. The data is generated, transmitted using MQTT, processed in Node-RED and displayed on a dashboard.

USC/UCT and upSkill Campus provided an opportunity to gain **practical exposure to IoT technologies and project development**. The internship helped me connect my ECE academic knowledge with practical applications and improve my technical and problem-solving skills.



I gained practical knowledge of **IoT, Node-RED, MQTT and JSON** and learned how to develop and troubleshoot an IoT monitoring system. The internship improved my **technical, problem-solving and project-management skills** and gave me valuable industry exposure.

I sincerely thank **my brother, Shivam Yadav**, for his valuable guidance and support throughout my project. I am also thankful to my **teachers, mentors, upSkill Campus/UCT team, friends and family** for their direct or indirect support and encouragement.

"Keep learning, stay consistent and focus on practical knowledge. Do not be afraid of mistakes, because every mistake is an opportunity to learn and improve. Make the most of every internship and project opportunity."

2. Introduction

2.1. About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

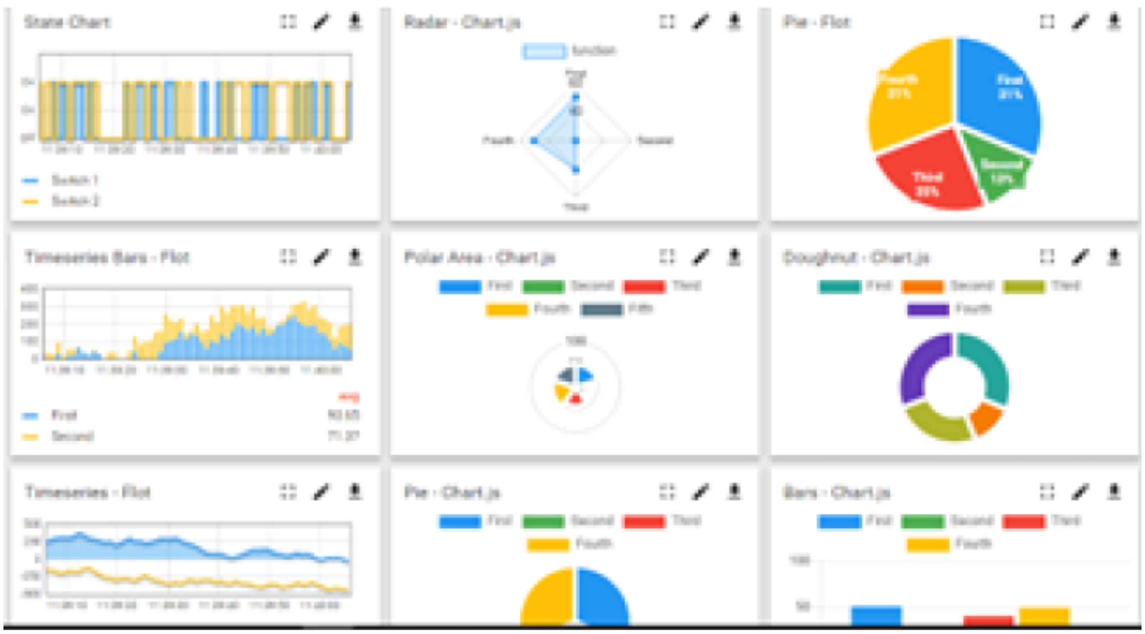
i. UCT IoT Platform (uct Insight)

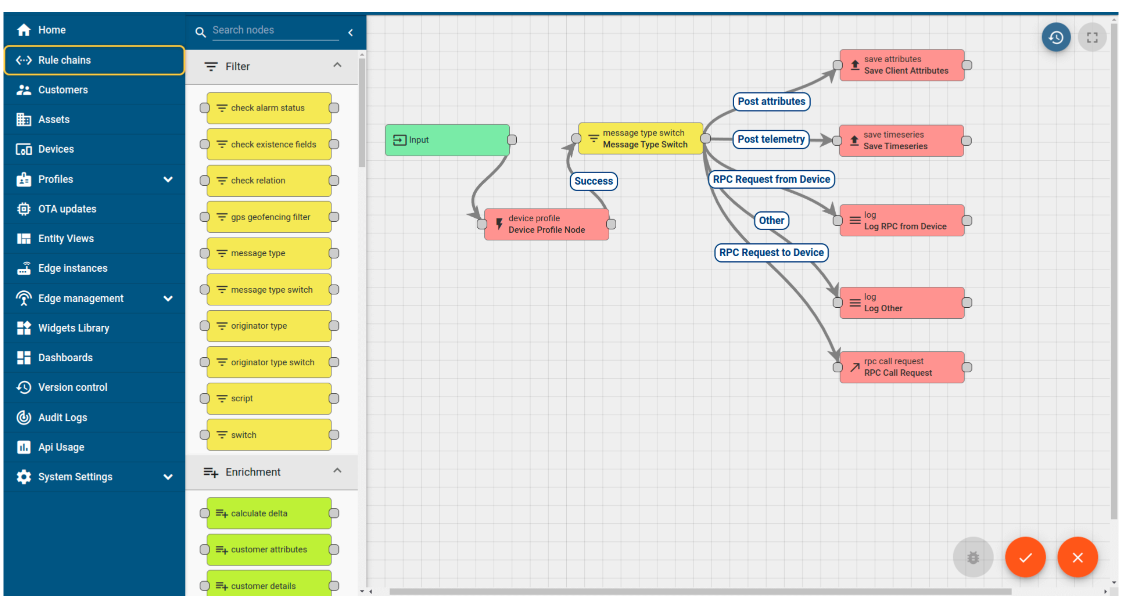
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

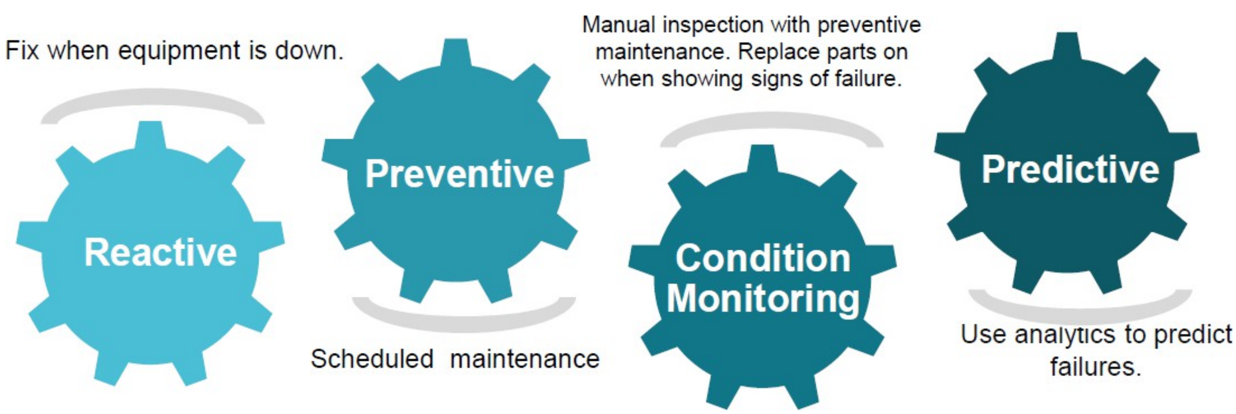
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

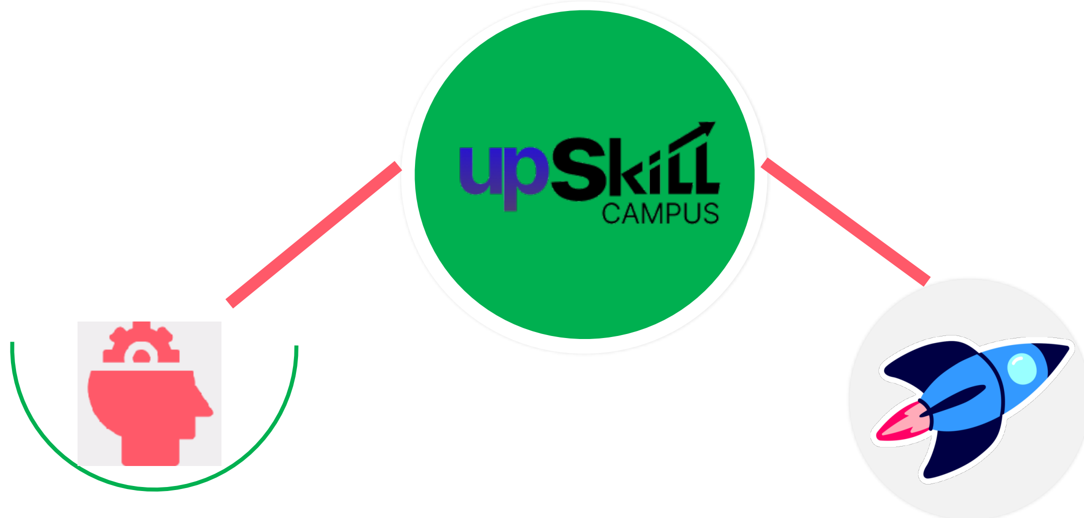
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2. About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

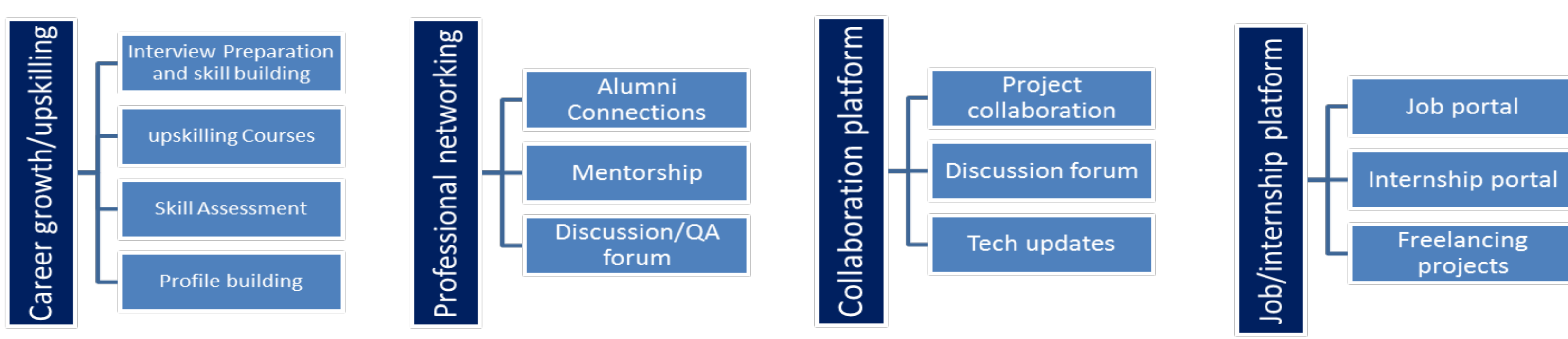
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/



2.3. The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4. Objectives of this Internship program

The objective for this internship program was to

☛get practical experience of working in the industry.

☛to solve real world problems.

☛to have improved job prospects.

☛to have Improved understanding of our field and its applications.

☛to have Personal growth like better communication and problem solving.

2.5. Reference

1. UpSkill Campus, Online Internship Program in Embedded and IoT, Internship Learning Resources
2. UpSkill Campus, *YouTube Channel*, IoT and Internship Learning Videos.
3. Node-RED, Official Documentation and User Guide, Node-RED Foundation.

2.6. Glossary

Terms	Acronym
Internet of Things	IOT
MessageQueuing Telemet Transport	MQTT
Javascript Object Notation	JSON
User Interface	UI

3. Problem Statement

Weather Monitoring System Using Node-RED and MQTT (Without Hardware)

Weather conditions such as **temperature and humidity** are important parameters for monitoring environmental conditions. In a conventional weather monitoring system, physical sensors are used to collect environmental data, and the collected information is then transmitted, processed and displayed through a monitoring application. However, working with physical sensors requires hardware components, wiring, configuration and additional testing.

The problem addressed in this project is to develop a **software-based weather monitoring system without using physical hardware**. The purpose is to understand and demonstrate the complete IoT monitoring process, starting from data generation and communication to processing and dashboard visualization.

In the proposed system, **simulated temperature, humidity and weather-status data** are generated instead of collecting data from physical sensors. The generated data is structured in **JSON format** and communicated using the **MQTT protocol**. MQTT provides a lightweight method for transferring messages between different parts of the IoT system.

Node-RED is used as the main development and processing platform. MQTT In and MQTT Out nodes are used for receiving and transmitting data, while Function nodes are used for processing the received information. Debug nodes are used to monitor the data flow and identify errors or incorrect data formats. Finally, the processed information is displayed on a **Node-RED dashboard** using suitable dashboard widgets.

The main challenge in the project is to establish a proper flow of data between the different components. The data must be generated in the correct format, transmitted through the appropriate MQTT topic, received correctly, processed by Node-RED and finally displayed on the dashboard. During development, issues such as incorrect message formats and undefined dashboard values can occur. These problems need to be identified and solved through proper debugging and testing.

The project therefore focuses on understanding the following complete IoT workflow:

Data Generation → JSON Formatting → MQTT Out → MQTT Topic → MQTT In → Function/ Processing → Dashboard

The project provides a basic foundation for understanding how an IoT monitoring system works without requiring physical sensors. Once the software-based system is successfully implemented, it can be extended in the future by replacing simulated data with **real sensor data** using suitable sensors and a microcontroller.

3.1.Existing and Proposed solution

1. **Summary of Existing Solutions and Their Limitations**

Existing weather-monitoring systems generally use **physical sensors, microcontrollers and IoT communication protocols** to collect and transmit environmental data. MQTT is commonly suitable for IoT communication because it is lightweight and uses a publish/subscribe model.

- Limitations:

  Requires physical sensors and hardware.
  Hardware setup and calibration increase complexity.
  Initial testing and debugging can be time-consuming.
  Hardware availability can limit experimentation.

2. Proposed Solution

I developed a **Weather Monitoring System using Node-RED and MQTT without hardware**. The system generates simulated **temperature, humidity and weather-status data**, sends it through MQTT, processes it using Node-RED and displays the results on a dashboard.

Node-RED provides a flow-based environment for connecting and processing different nodes.

Flow:
Simulated Data → JSON → MQTT → Node-RED Processing → Dashboard

3. Value Addition Planned

- Integrating real temperature and humidity sensors.
- Implementing threshold-based alerts.
- Extending the system for remote/cloud monitoring.

3.2. Code submission (Github link): https://github.com/Kunal30-08/upskillcampus/blob/main/WeatherMonitoringSystem.json

3.3. Report submission (Github link): https://github.com/Kunal30-08/upskillcampus/blob/main/WeatherMonitoringSystem_Kunal_USC_UCT.pdf

4. Proposed Design/ Model

The proposed system is a software-based **Weather Monitoring System using Node-RED and MQTT without physical hardware**. Simulated weather data is generated and processed through a sequence of Node-RED nodes. The data contains parameters such as temperature, humidity and weather status.The generated data is converted into **JSON format** and communicated using MQTT. The received data is then processed and separated into individual parameters. Finally, the processed values are displayed on the Node-RED Weather Dashboard using gauges, graphs and status indicators.

4.1. High Level Diagram

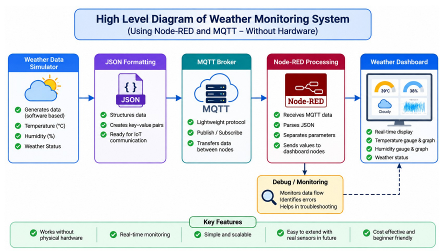
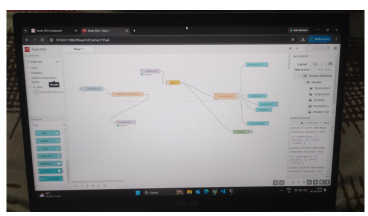
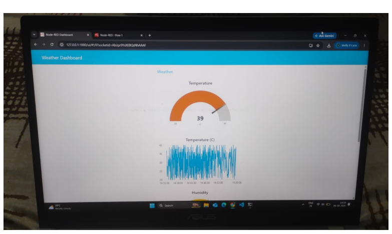


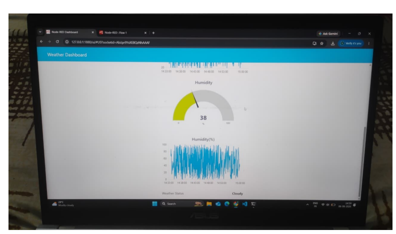
Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

4.2. Low Level Diagram



4.3. Interfaces





5. Performance Test

Constraints:
No physical sensors were used. MQTT communication and data-format issues were the main constraints.

Handling:
Debug nodes were used to identify errors and verify the data flow.

Test Result:
Temperature, humidity and weather-status data were successfully generated, transmitted, processed and displayed.

5.1. Test Plan/ Test Cases

- Data generation — **Pass**
- JSON formatting — **Pass**
- MQTT communication — **Pass**
- Data processing — **Pass**
- Dashboard display — **Pass**

5.2. Test Procedure

Generate data → Convert to JSON → Send through MQTT → Process in Node-RED → Display on dashboard.

5.3. Performance Outcome

The system successfully demonstrated the complete IoT data flow from **data generation to dashboard visualization**. Temperature, humidity and weather status were displayed successfully. The project met its basic objectives and provided a functional software-based weather monitoring prototype.

Note: Since this is a **without-hardware prototype**, parameters such as sensor accuracy, power consumption and hardware durability were not measured.

6. My learnings

In the first week, I focused on understanding the **company, internship program and its objectives**. I learned about the organization, the internship structure and the overall approach followed during the program. This helped me understand how the internship would contribute to my technical learning and project development.

In the second week, I learned about **IoT devices and the evolution of the Internet of Things** through the videos provided in the internship course. I also watched the podcast featuring **Mr. Arun Kumar Singh**, which provided valuable insights into the growth of IoT and optical fibre networking.During this week, I explored different project ideas and finalized my project as **"Weather Monitoring System (Without Hardware)"**. I also started the initial phase of the project and planned the basic requirements.

In the third week, I studied **IoT Deployment and IoT Deployment Challenges**. I learned about security, design, deployment and scalability challenges in IoT systems. I also learned about **embedded systems** and their role in IoT applications.Along with the theoretical learning, I continued my project work. I defined the **problem statement**, created a **Node-RED dashboard** and successfully displayed simulated **Temperature, Humidity and Weather Status** values. I also started learning the basics of **MQTT** and understood its role in IoT communication.

In the fourth week, I improved my basic knowledge of **electrical and electronic concepts**, including current, voltage, resistance, transistors, capacitors, inductors, diodes, Ohm's law and the basic use of a volt-ohm meter.

In the fifth week, I focused on **testing and improving the complete project flow**. I checked the communication between the different Node-RED nodes and verified that the simulated weather data was correctly transferred through MQTT and displayed on the dashboard.

Overall, the internship gave me practical exposure to **IoT, Node-RED, MQTT, JSON, dashboard development, basic electronics and troubleshooting**. I learned how to take a project from understanding the problem and planning a solution to implementation, testing and documentation.
The internship also improved my **problem-solving, debugging, technical understanding and project-management skills**. As an ECE student, this experience helped me connect my academic knowledge with practical IoT applications and provided a strong foundation for my future career in **IoT, embedded systems, electronics and related technologies**.

7. Future work scope

The current project successfully demonstrates a software-based weather monitoring system using Node-RED and MQTT. Due to the limited internship duration, some advanced features could not be implemented. These can be taken up as future improvements:

- **Real Sensor Integration:** Replace simulated data with real temperature and humidity sensors connected to a microcontroller.
- **Data Storage:** Store weather readings in a database for future analysis.
- **Historical Graphs:** Add detailed graphs to analyze temperature and humidity trends over time.
- **Alert System:** Generate alerts when temperature or humidity crosses predefined limits.
- **Cloud Integration:** Connect the system to a cloud platform for remote monitoring.
- **Mobile Access:** Develop a mobile-friendly interface for accessing weather data remotely.
- **Security:** Implement secure MQTT communication and user authentication.
- **Advanced Weather Parameters:** Add parameters such as air pressure, rainfall and air quality.
- **Predictive Analysis:** Use collected historical data to identify weather patterns and make basic predictions.
- **Hardware Expansion:** Integrate additional sensors and make the system suitable for real-world deployment.

These improvements can make the project more **accurate, scalable, secure and suitable for real-world IoT applications**.